*Recommendations Report*

1. **Recommended Model: Random Forest**

In evaluating the model results, I would suggest Random Forest as the ideal model for the StadioChoice trial to paid conversion.

Just looking at accuracy alone can be misleading in an imbalanced data set but the F1 score of 51.3% against the 46% on Logistic Regression proves that Random Forest achieves a much better overall balance between identifying actual converters and not wasting effort.

For the STADIOChoice this will be a big consideration because acquiring a streaming customer can cost R355 while the streaming division is losing money. Spamming trial users who wont even have an intention of staying would be wasting marketing budget and can push customer away by annoying. So Random Forest higher precision of 44.4% means the marketing campaigns are then sent to people who are actually likely to convert.

A factor to consider despite this recommendation is Logistic Regression has a slightly higher recall of 65% against 61% which then means it catches on more total overall converters. This would makes sense if ever StadioChoice cares soley on getting as many sign ups possible no matter the spending.

2. **How the model can be improved**

Because Random Forest is essentially a tree based model, it already navigates with the nonlinear relationships and feature interactions. Instead of just adding basic interaction terms there could be different ways to improve a Random Forest model performance:

Threshold optimisation: Rather than shifting the decision boundaries soley to boost recall at the cost of precision, the threshold should be selecting by being able to plot recall curves against the StadioChoice business costs, so weighing R355 acquisition cost against the cost of a wasted campaign.

Feature Selection: To consider applying recursive feature elimination or structured featured procedures would be able to isolate high impact behavioural factors.

Class resampling: If in the first place there was no resampling implemented in the training pipeline, applying techniques such as SMOTE would be ideal because they would balance the minority class aspect during the training phase.

3. **How will the selected model need to be adapted to work with SS1 data**

Adapting this model approach from the bank marketing data set to StadioChoice will need changes in the way the data is pre processed, strategy taken for missing values, and ensuring there aren't data leakages.

The Bank marketing data set used customer demographic fields such as job, marital status, education, etc. and macroeconomic factors. While SS1 data looked at combined trial period behaviour such as total watch time, searchers made, searches with no results, device, genres watched, sign up platform and trial plan. The model would be kept on these fields while still having missing behavioural values still set to zero when a customer did nothing.

To then prevent data leakage only the information available during the trial can be used as the input. The date of conversion and early churn would happen after the trial and should be excluded, converted to paid subscription is the target and then Account ID would just be the identifier. This is also the reason to drop the duration column in the Bank marketing data.

Lastly, StadioChoice converts 33% of the trial users compared to 11% shown in the Bank Marketing data set.

4. **Do your results align with the related literature from Part A?**

My results align well to certain parts of the literature I went through but there were also parts that had a different approach on my side.

(Moro et al., 2014) was evaluating multiple models(Logistic Regression, Decision Tree, Neural Network and SVM) for bank telemarketing success. Although they did not use Random Forest, my approach aligns with their fundamental strategy of comparing multiple modelling techniques rather than relying on one which is what I did in comparing two models.

They also showed the importance of rigorous feature selection( reducing 150 features to 22) and having a time based train test split to show a real world campaign progression

Lower F1 score (Different from the literature) : (Kostovska and Gjoreski, 2025) had a high F1 of 0.87 when using tree models on the trial data and my results different by having a lower F1 scoring of 0.513 from my Random Forest. I would assume this difference is because they had more session by session data and my data was using more broad field terming and removed duration/leakage columns.

The similar ROC AUC score (Matched the literature): The (Jagelid and Movin, 2017) used AUC as a key metric for imbalanced data and found that Logistic Regression performance comparably to Gradient Boosting. This was a similar finding to my results because the gap between the Random Forest of 80.9% and Logistic Regression of 80% was a very small difference.

**References**

JAGELID, T. & MOVIN, J. 2017. *From Free to Fee: Converting Freemium Users into Premium Subscribers*. Bachelor's thesis, Lund University.

KOSTOVSKA, A. & GJORESKI, F. 2025. Predicting Subscription Conversion from Free Trial Usage Data. *arXiv preprint*.

MORO, S., CORTEZ, P. & RITA, P. 2014. A data-driven approach to predict the success of bank telemarketing. *Decision Support Systems,* 62**,** 22–31.